ASSIGNMENT 8: OPTIMIZE ANYTHING

The 14-Day Vibe-Coding Challenge — Technical Report

Course: Optimization Techniques | **Author**: AI-Optimization-Researcher | **Date**: October 2026 | **Status**: All 10 Tracks A-J Complete (10/10 Tests PASS)

Executive Summary

This report presents the complete empirical results for the 14-Day Optimization Challenge. Ten compulsory tracks were implemented, benchmarked, and verified using documented AI-assisted (vibe) coding. Across all domains—from low-level OS kernels and compiler backends to smart grid scheduling, relational databases, and quantum optimization—optimization demonstrated significant measurable gains while preserving 100% semantic correctness.

Ten-Track Experimental Results Summary

Track	Domain / Problem	Optimization Method	Measured Improvement	Correctness
Track A	Continuous Landscapes	Adam / Newton-CG / HVP	Quadratic convergence, saddle escape	PASS (<1e-10 err)
Track B	Discrete Knapsack & QUBO	ILP (HiGHS) / B&B; / SA	Optimal Cut 8.0 on 6-node graph	PASS (Exact)
Track C	Quantum Max-Cut	QAOA (p=2) / SPSA / VQE	Matches classical optimal cut (8.0)	PASS (Bitstring 010101)
Track D	Submodular & Parallel	Greedy / Work Stealing / FedAvg	98.2% of exact optimum (bound 63.2%)	PASS (0 violations)
Track E	Commercial EV Fleet	Mixed-Integer LP (HiGHS)	77.5% cost savings ($5.1k -> $1.1k)	PASS (0 overloads)
Track F	Compiler Optimization	4 Passes + ILP Register Allocation	5.0x cycle speedup, -80% code size	PASS (Bit-identical)
Track G	xv6 Operating System	MLFQ Scheduler + Dynamic Slices	-57% context switches, 4x response	PASS (Zero deadlock)
Track H	DBMS Workload (TPC-H)	Knapsack Index Selection + 2Q	2.15x query speedup within 400KB	PASS (Bit-identical)
Track I	Quantum Algorithms & QML	7 Algos + VQC vs Classical MLP	VQC matched MLP at 73.8% acc	PASS (7/7 Algos)
Track J	Software Engineering	ILP CI Test Minimization	2.80x faster CI (487ms -> 174ms)	PASS (86.5% cov)

Key Technical Highlights

Track E (Smart Fleet Charging): Formulating the fleet charging problem as an MILP shifted charging from on-peak tariff periods ($0.38/kWh) to off-peak periods ($0.08/kWh). This reduced peak substation demand from 363 kW to 68 kW, completely eliminating 2 hours of transformer overloads and slashing daily electricity costs by 77.5%.

Track F (Compiler Passes): Combining Constant Folding, Dead Code Elimination, and Common Subexpression Elimination reduced TAC instructions by 80% on Program 1, executing in 2 cycles instead of 10 while maintaining 100% verified output equivalence.

Track G (xv6 OS Kernel): Replacing Round-Robin with an MLFQ scheduler reduced context switches from 130 to 56 and accelerated interactive process response times from 4 ticks to 1 tick.

Track H (DBMS Index Selection): Formulated index selection as a 0-1 Knapsack problem under a 400 KB disk limit. Dynamic Programming and ILP identified the optimal index set (idx_ord_custkey, idx_cust_mktseg, idx_li_suppkey), delivering a 2.15x speedup over sequential scans.

Vibe Coding & AI Failure Log Summary

Three substantive AI failures were identified, analyzed, and corrected during development:
1. **Compiler DCE**: Naive AI liveness analysis stripped print and memory writes with side effects; resolved by whitelisting essential opcodes.
2. **xv6 State Machine**: AI omitted burst_idx incrementing on ZOMBIE transition, causing an apparent kernel hang; resolved by atomic state updates.
3. **QAOA Gradient Shift**: Simultaneous parameter-shift across multi-gate shared parameters corrupted gradients; resolved by SPSA and finite differences.